

Animals and Humans								
	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I know that humans and other animals grow and change over time. I know that different animals live in different places and eat different foods. I can recognise that some animals are active during the day and some are active at night. I can identify some similarities and differences between familiar animals. I can observe animals in the natural world and talk about what I	I can observe animals and talk about how they grow and change. I can talk about the life cycles of familiar animals. I can describe some of the things that animals, including humans, need to survive. I can identify similarities and differences between familiar animals. I can explore the natural world and make observations of animals. I can represent my	I can identify and name a variety of common animals, including fish, amphibians, reptiles, birds and mammals. I can identify and name a variety of common animals that are carnivores, herbivores and omnivores. I can describe and compare the structures of a variety of common animals, including fish, amphibians, reptiles, birds, mammals and pets. I can identify, name, draw and label the basic	I can notice that animals, including humans, have offspring which grow into adults. I can describe the basic needs of animals, including humans, for survival: water, food and air. I can describe the importance of exercise for humans. I can describe the importance of eating the right amounts of different types of food.	I can identify that animals, including humans, need the right types and amounts of nutrition. I know that animals cannot make their own food and get the nutrition they need from what they eat. I can identify that humans and some other animals have skeletons and muscles. I can explain how skeletons provide support, protection and movement. I can explain how muscles work with the skeleton to enable	I can describe the simple functions of the basic parts of the human digestive system. I can identify the different types of human teeth and describe their simple functions. I can construct and interpret a variety of food chains. I can identify producers, consumers, predators and prey within food chains.	I can describe the changes that take place as humans develop from birth to old age. I can describe the main stages of the human life cycle. I can describe changes associated with puberty. I can understand that reproduction is part of the human life cycle. I can compare aspects of human development with the life cycles of other animals.	I can identify and name the main parts of the human circulatory system. I can describe the functions of the heart, blood vessels and blood. I can describe how nutrients and water are transported within animals, including humans. I can recognise and explain the impact of diet and exercise on the way our bodies function. I can recognise the impact that drugs and other lifestyle choices can have on the way our bodies

	notice. I can represent my observations through talk, drawing and other forms of recording.	observations through talk, drawing and other forms of recording. I can show care and concern for living things and the environment.	parts of the human body. I can identify which parts of the body are associated with each of the senses.	I can describe the importance of good hygiene for humans.	movement.			function.
Vocabulary	Human, grow, young, old, same, different, food, habitat, day, night, nocturnal, diurnal, observe	Human, grow, young, adult, life cycle, survive, food, water, air, habitat, alive, dead, same, different, similar, observe, change	Human, fish, amphibian, reptile, bird, mammal, carnivore, herbivore, omnivore, body, head, neck, arms, legs, eyes, ears, nose, mouth, skin, wings, fins, tail, gills, senses, sight, hearing, smell, taste, touch, similarities, differences	Water, air, food, offspring, adult, baby, young, grow, growth, exercise, physical activity, healthy, diet, hygiene, clean, nutrition, life cycle	Nutrients, balanced diet, carbohydrates, protein, fat, fibre, skeleton, bones, muscles, joints, ribs, skull, backbone, spine, support, protection, movement, vertebrate, invertebrate	Digestive system, mouth, oesophagus, stomach, small intestine, large intestine, rectum, anus, saliva, teeth, incisor, canine, premolar, molar, cutting, tearing, grinding, food chain, producer, consumer, predator, prey	Growth, life cycle, baby, toddler, child, teenager, adult, elderly, ageing, puberty, reproduction, reproduce, embryo, womb, pregnancy, gestation	Heart, blood, blood vessels, artery, vein, capillary, lungs, oxygen, carbon dioxide, nutrients, oxygenated, deoxygenated, pump, circulation, transport, diet, exercise, drugs, lifestyle

Living things & their habitats								
	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can observe animals and plants in the natural world and talk about what I notice. I can describe some things that I see, hear and feel when exploring outside. I can identify similarities, differences, patterns and changes in the natural world. I can begin to understand that my actions can affect living things and the environment. I can show care for living things and the natural environment.	I can explore and make observations of different habitats. I can identify some animals and plants found in familiar habitats. I can talk about why particular animals are suited to the places in which they live. I can describe features of my immediate environment. I can recognise that different environments have different features. I can ask questions about living things, habitats and the natural world.	Not covered in this year group.	I can explore and compare the differences between things that are living, dead and have never been alive. I can identify that most living things live in habitats to which they are suited. I can describe how different habitats provide for the basic needs of different kinds of animals and plants. I can describe how animals and plants within a habitat depend on each other. I can identify and name a variety of plants and animals in their habitats, including microhabitats.	Not covered in this year group.	I can recognise that living things can be grouped in a variety of ways. I can explore and use classification keys to group, identify and name a variety of living things. I can classify living things found in the local and wider environment using observable characteristics. I can recognise that environments can change. I can explain how environmental change can sometimes pose dangers to living things. I can construct and interpret food chains, identifying producers,	I can describe the differences in the life cycles of a mammal, amphibian, insect and bird. I can describe the life process of reproduction in some plants. I can describe the life process of reproduction in some animals. I can compare the life cycles of different groups of animals. I can identify similarities and differences between different life cycles.	I can describe how living things are classified into broad groups according to common observable characteristics. I can classify microorganisms, plants and animals using similarities and differences in their characteristics. I can use classification systems and keys to identify living things. I can give reasons for classifying plants and animals based on specific characteristics. I can explain and justify my classification decisions using scientific evidence and vocabulary.

				I can describe how animals obtain their food from plants and other animals. I can construct a simple food chain and identify different sources of food.		consumers, predators and prey.		
Vocabulary	Animal, plant, environment, nature, grow, change, same, different, similar, observe, notice, care	Animal, plant, living, environment, natural world, hot, cold, desert, rainforest, woodland, pond, river, sea, park, same, different, similar, observe, describe, explain	Not covered in this year group.	Microhabitat, living, dead, never alive, animal, plant, suited, survival, needs, depend, food, shelter, woodland, pond, ocean, rainforest, desert, food chain, source, producer, consumer	Not covered in this year group.	Classify, classification key, group, characteristics, organism, living thing, environment, habitat, environmental change, producer, consumer, predator, prey, food chain	Life cycle, reproduction, reproduce, mammal, amphibian, insect, bird, egg, offspring, young, adult, metamorphosis, growth, development, pollination, fertilisation, seed, dispersal	Classification, classify, classification key, characteristics, similarities, differences, organism, microorganism, plant, animal, vertebrate, invertebrate, group, subgroup, criteria

Plants								
	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can observe plants in the natural world and talk about what I notice.	I can make observations of plants and describe what I notice.	I can identify and name a variety of common wild plants.	I can observe and describe how seeds and bulbs grow into mature plants.	I can identify and describe the functions of the roots, stem or trunk, leaves and flowers of flowering plants.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.
	I can identify some familiar plants and parts of plants.	I can identify and describe some parts of familiar plants.	I can identify and name a variety of common garden plants.	I can describe the stages of growth from a seed or bulb to a mature plant.	I can explore the requirements of plants for life and growth, including air, light, water, nutrients from the soil and room to grow.			
	I can notice that plants grow and change over time.	I can describe how plants grow and change over time.	I can identify and name deciduous and evergreen trees.	I can investigate and describe how plants need water, light and a suitable temperature to grow and stay healthy.	I can recognise that the requirements for growth can vary between different plants.			
	I can begin to recognise that plants need water and light to grow.	I can recognise that plants need certain things to grow.	I can identify and describe the basic structure of a variety of common flowering plants.	I can recognise that different conditions can affect plant growth.	I can investigate how water is transported within plants.			
	I can identify some similarities and differences between plants.	I can ask questions about plants and the changes I observe.	I can identify and describe the basic structure of trees.		I can explain the role flowers play in the life cycle of flowering plants.			
	I can show care for plants and the natural environment.	I can compare plants and identify similarities and differences.			I can describe the processes of			

					pollination, seed formation and seed dispersal.			
Vocabulary	Flower, tree, leaf, leaves, stem, roots, seed, soil, water, sunlight, grow, change, same, different, living, care	Plant, flower, tree, leaf, leaves, stem, roots, seed, bud, soil, water, sunlight, grow, growth, change, living, same, different, similar, observe, describe	Plant, wild plant, garden plant, flowering plant, tree, deciduous, evergreen, roots, stem, trunk, branch, leaf, leaves, flower, blossom, petals, fruit, seed, bulb	Seed, bulb, germination, seedling, mature plant, growth, roots, shoot, leaves, water, light, dark, warmth, temperature, healthy, unhealthy, conditions	Function, flowering plant, roots, root hair, stem, trunk, leaves, flower, air, light, water, nutrients, soil, growth, transport, life cycle, pollination, pollen, seed formation, seed dispersal, reproduce, stamen, anther, filament, stigma, style, ovary, fruit	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.

Electricity

	Nursery/Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	I can identify common appliances that run on electricity. I can construct a simple series electrical circuit. I can identify and name the basic parts of a circuit, including cells, wires, bulbs, switches and buzzers. I can identify whether a lamp will light in a simple series circuit by recognising whether the circuit forms a complete loop with a cell. I can recognise that a switch opens and closes a circuit. I can explain how opening and closing a switch affects whether a lamp lights. I can identify some common electrical conductors and	Not covered in this year group.	I can associate the brightness of a lamp with the number and voltage of cells used in a circuit. I can associate the volume of a buzzer with the number and voltage of cells used in a circuit. I can compare variations in how electrical components function. I can give reasons for variations in the brightness of bulbs, the loudness of buzzers and the operation of switches. I can use recognised electrical symbols when representing a simple circuit in a diagram. I can construct and interpret circuit diagrams using recognised electrical

					insulators. I can recognise that metals are generally good electrical conductors.		symbols.
Vocabulary	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Electricity, electrical, appliance, mains, cell, battery, wire, bulb, lamp, buzzer, motor, switch, circuit, series circuit, component, complete circuit, incomplete circuit, terminal, connect, disconnect, conductor, insulator, metal	Not covered in this year group.	Voltage, cell, battery, circuit, series circuit, component, lamp, bulb, buzzer, motor, switch, wire, terminal, brightness, volume, conductor, insulator, circuit diagram, electrical symbol

Materials								
	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can identify and name some familiar materials and natural objects. I can explore different materials through play and describe what I notice. I can describe some materials using simple words relating to their texture and appearance. I can notice that some materials can be changed by actions such as squashing, bending or mixing. I can explore how different materials can be used when building and constructing.	I can identify and name a range of familiar materials. I can describe and compare some properties of materials. I can begin to explain why particular materials are suitable for different purposes. I can explore what happens to different materials when they come into contact with water. I can investigate whether familiar objects float or sink. I can observe and describe changes in materials, including melting and freezing. I can use my observations to make simple	I can distinguish between an object and the material from which it is made. I can identify and name a variety of everyday materials, including wood, plastic, glass, metal, water and rock. I can describe the simple physical properties of a variety of everyday materials. I can compare and group materials based on their physical properties.	I can identify and compare the suitability of a variety of everyday materials for particular uses. I can describe properties that make particular materials suitable or unsuitable for a purpose. I can compare different materials and explain which would be most suitable for a particular use.	Not covered in this year group.	I can compare and group materials according to whether they are solids, liquids or gases. I can describe some of the properties of solids, liquids and gases. I can observe that some materials change state when they are heated or cooled. I can measure or research the temperature at which changes of state happen in degrees Celsius (°C). I can identify evaporation and condensation within the water cycle. I can explain how	I can compare and group everyday materials based on their properties, including hardness, solubility, transparency, electrical and thermal conductivity and response to magnets. I can recognise that some materials dissolve in liquids to form solutions. I can describe how a dissolved substance can be recovered from a solution. I can use my knowledge of solids, liquids and gases to decide how mixtures can be separated, including through filtering, sieving and evaporation.	Not covered in this year group.

		choices about which materials to use for a particular purpose.				temperature affects the rate of evaporation.	I can give reasons, based on evidence from comparative and fair tests, for the uses of particular everyday materials. I can demonstrate that dissolving, mixing and changes of state can be reversible changes. I can explain that some changes result in the formation of new materials and are not usually reversible. I can identify burning and the reaction between acid and bicarbonate of soda as examples of changes that form new materials.	
Vocabulary	Material, wood, plastic, paper, cardboard, fabric, natural, object, hard, soft, rough, smooth, heavy, light, build, change,	Material, property, wood, plastic, paper, cardboard, fabric, metal, hard, soft, strong, weak, rough, smooth, heavy, light,	Material, object, property, wood, plastic, glass, metal, water, rock, brick, fabric, hard, soft, rough, smooth, shiny, dull,	Suitability, material, property, purpose, wood, metal, plastic, glass, brick, rock, paper, cardboard, fabric, hard, soft,	Not covered in this year group.	State of matter, solid, liquid, gas, properties, temperature, degrees Celsius, heating, cooling, melting, freezing,	Properties, hardness, solubility, soluble, insoluble, transparency, transparent, opaque, electrical	Not covered in this year group.

	texture	waterproof, absorbent, float, sink, melt, freeze, change, compare	stiff, flexible, heavy, light, transparent, opaque, absorbent, waterproof	flexible, stiff, strong, weak, rough, smooth, transparent, opaque, waterproof, absorbent		melting point, freezing point, evaporation, evaporate, condensation, condense, water vapour, water cycle	conductor, electrical insulator, thermal conductor, thermal insulator, magnetic, non-magnetic, dissolve, solution, mixture, substance, filtering, sieving, evaporation, reversible change, irreversible change, burning, reaction	
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Forces

	EYFS (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can explore how objects move when I push and pull them. I can notice that different actions can make objects move, stop or change direction. I can explore how objects move during play and talk about what I notice.	I can explore and describe how pushes and pulls affect the movement of objects. I can compare how objects move in different ways and on different surfaces. I can explore how the strength of a push or pull can affect an object's movement. I can ask questions and talk about why objects move, stop or change direction.	Not covered in this year group.	Not covered in this year group.	I can compare how objects move on different surfaces. I can recognise that some forces require contact between two objects. I can recognise that magnetic forces can act at a distance. I can observe how magnets attract or repel each other. I can investigate which materials are attracted to magnets. I can compare and group materials according to whether they are attracted to a magnet. I can identify some magnetic	Not covered in this year group.	I can explain that unsupported objects fall towards the Earth because of the force of gravity acting between the Earth and the falling object. I can identify the effects of friction between moving surfaces. I can identify the effects of air resistance on moving objects. I can identify the effects of water resistance on moving objects. I can explain how resistance can affect the movement of an object. I can recognise that some mechanisms,	Not covered in this year group.

					materials. I can describe magnets as having two poles. I can predict whether two magnets will attract or repel each other depending on which poles are facing.		including levers, pulleys and gears, allow a smaller force to have a greater effect.	
Vocabulary	Push, pull, move, stop, fast, slow, direction, object	Force, push, pull, movement, direction, fast, slow, stop, surface, strong, weak	Not covered in this year group.	Not covered in this year group.	Force, contact force, non-contact force, push, pull, movement, surface, friction, magnet, magnetic, non-magnetic, attract, repel, pole, north pole, south pole	Not covered in this year group.	Gravity, force, weight, mass, friction, air resistance, water resistance, drag, surface area, streamlined, mechanism, lever, pivot, fulcrum, pulley, gear, cog	Not covered in this year group.

Light

	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can explore light and dark through play. I can explore what happens when I shine a torch on different objects and surfaces. I can notice that some objects and places are brighter or darker than others. I can begin to notice that objects can make shadows when light is blocked. I can talk about what I notice when exploring light and dark.	I know that the Sun is a natural source of light. I can identify some other sources of light. I can explore how shadows are made when light is blocked by an object. I can observe that different materials allow different amounts of light to pass through them. I can explore how light can appear different when it passes through coloured materials. I can describe what I notice when exploring light, darkness and shadows.	Not covered in this year group.	Not covered in this year group.	I can recognise that we need light in order to see things. I can recognise that darkness is the absence of light. I can notice that light is reflected from surfaces. I can recognise that light from the Sun can be dangerous and identify ways to protect our eyes. I can recognise that shadows are formed when light from a light source is blocked by an opaque object. I can investigate and identify patterns in the way the size of shadows changes.	Not covered in this year group.	Not covered in this year group.	I can recognise that light appears to travel in straight lines. I can use the idea that light travels in straight lines to explain how we see objects. I can explain that objects are seen because they give out or reflect light into the eye. I can explain that we see things because light travels from a light source to our eyes, or from a light source to an object and then to our eyes. I can use the idea that light travels in straight lines to explain how shadows are formed.

								I can explain why shadows have the same shape as the objects that cast them.
Vocabulary	Light, dark, bright, torch, shine, shadow, sunlight, day, night	Light source, light, dark, bright, dim, Sun, sunlight, torch, shadow, block, material, transparent, translucent, opaque	Not covered in this year group.	Not covered in this year group.	Light, light source, dark, darkness, shadow, reflection, reflect, reflective, surface, mirror, bright, dim, opaque, transparent, translucent, Sun, ultraviolet, protection	Not covered in this year group.	Not covered in this year group.	Light, light source, ray, straight line, reflection, reflect, eye, object, shadow, opaque, transparent, translucent, mirror, ray diagram

Sound							
	Nursery/Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	I can identify how sounds are made and associate some sounds with something vibrating. I can recognise that vibrations from sounds travel through a medium to the ear. I can find patterns between the pitch of a sound and features of the object that produced it. I can find patterns between the volume of a sound and the strength of the vibrations that produced it. I can recognise that sounds become fainter as the distance from the sound source increases. I can use my understanding of vibration, pitch and volume to explain	Not covered in this year group.	Not covered in this year group.

					differences between sounds.		
Vocabulary	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Sound, sound source, vibration, vibrate, medium, ear, pitch, high, low, volume, loud, quiet, fainter, distance, strength, travel	Not covered in this year group.	Not covered in this year group.

Seasonal Change

	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can notice and talk about changes in the weather. I can notice changes in the natural world around me at different times of the year. I can explore the outdoor environment in different weather conditions. I can begin to recognise that some types of weather are associated with different times of the year.	I can observe and describe changes in the natural world across the seasons. I can recognise and name the four seasons. I can describe some types of weather associated with different seasons. I can notice and describe how the environment changes through the year. I can begin to notice that the amount of daylight changes at different times of the year.	I can name and identify the four seasons: spring, summer, autumn and winter. I can observe changes across the four seasons. I can observe and describe weather associated with each season. I can observe and describe how day length varies across the seasons. I can record and compare seasonal and weather changes over time.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.
Vocabulary	Weather, season, change, Sun, rain, cloud, wind, snow, hot, cold, warm, day, outside	Season, spring, summer, autumn, winter, weather, temperature, sunlight, daylight,	Season, spring, summer, autumn, winter, weather, temperature, day length, daylight,	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.

		rain, snow, frost, cloud, wind, hot, warm, cool, cold, change	sunrise, sunset, Sun, rain, snow, frost, ice, cloud, sunny, cloudy, windy, stormy, hot, warm, cool, cold, change, observe					
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Rocks

	Nursery/Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	I can compare and group different kinds of rocks based on their appearance and simple physical properties. I can describe some of the properties of different types of rock. I can describe in simple terms how fossils are formed when living things become trapped within rock. I can recognise that soils are made from rocks and organic matter. I can compare different types of soil based on their properties.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.
Vocabulary	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Rock, properties, sandstone, granite, marble, pumice, crystal, hard, soft,	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.

				absorbent, fossil, fossilisation, remains, sediment, soil, organic matter, clay, sand, loam, peat			
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Evolution and Inheritance

	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can notice that living things grow and change over time. I can identify some similarities and differences between living things. I can notice that young animals, including humans, grow into adults.	I can describe some ways that living things grow and change over time. I can identify similarities and differences between animals, plants and people. I can recognise that young animals are similar to, but not exactly the same as, their parents.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	I can recognise that living things have changed over time. I can recognise that fossils provide information about living things that inhabited the Earth millions of years ago. I can recognise that living things produce offspring of the same kind, but that offspring normally vary and are not identical to their parents. I can identify how animals and plants are adapted to suit their environment in different ways. I can explain how adaptations can help living things to survive in their environments. I can understand

								that adaptation may lead to evolution over long periods of time.
Vocabulary	Change, grow, young, adult, same, different, similar, animal, human	Similarities, differences, change, grow, young, adult, parent, offspring, same, different, similar	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Evolution, inheritance, inherited, offspring, variation, characteristics, adaptation, adapted, environment, survival, fossil, fossil record, ancestor, species, generation

Earth & Space

	Nursery/Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	I can describe the movement of the Earth and other planets relative to the Sun in the solar system. I can describe the movement of the Moon relative to the Earth. I can describe the Sun, Earth and Moon as approximately spherical bodies. I can explain that the Earth rotates on its axis. I can use the Earth's rotation to explain day and night. I can use the Earth's rotation to explain the apparent movement of the Sun across the sky.	Not covered in this year group.
Vocabulary	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Not covered in this year group.	Solar system, Sun, star, Earth, planet,	Not covered in this year group.

						Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune, Moon, spherical, orbit, rotation, rotate, axis, day, night, sunrise, sunset	
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Working Scientifically (Observation over time, Pattern seeking, Identifying, classifying and grouping, Comparative and fair testing, Research using secondary sources)								
	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	Planning							
	I know that I can find things out by looking closely, listening and talking to others. I can ask simple questions about things I notice. I can explore things to find out what happens.	I can ask questions about the world around me. I know that I can observe, explore and test things to help answer my questions. I can suggest simple ways to find something out.	I can ask simple questions when prompted. I can suggest ways of answering a question.	I can ask simple scientific questions. I can recognise that questions can be answered in different ways. I can choose a simple way to investigate a question.	I can ask relevant scientific questions. I can set up simple practical enquiries. I can set up comparative and fair tests with support.	I can ask relevant scientific questions. I can choose an appropriate type of enquiry to answer a question. I can independently set up practical enquiries, comparative tests and fair tests.	I can plan different types of scientific enquiry to answer questions with some prompting. I can identify variables within an investigation. I can recognise when variables need to be controlled.	I can independently plan different types of scientific enquiry to answer questions. I can identify and control variables where necessary. I can explain why variables need to be controlled to make an investigation valid.
	Conducting Experiments							
	I can explore what happens when I use different objects, materials and equipment. I can observe what happens when I pour, fill, build, balance, push, pull	I can carry out simple practical explorations. I can use simple equipment to observe things closely. I can compare what happens	I can make relevant observations. I can use simple equipment. I can carry out simple tests with support.	I can observe closely using simple equipment. I can perform simple tests. I can use simple measurements and observations to	I can make systematic and careful observations. I can use a range of simple scientific equipment. I can take measurements	I can make systematic and careful observations. I can use a range of equipment, including thermometers and data loggers where	I can select appropriate equipment with some prompting. I can take increasingly accurate and precise measurements using standard	I can independently select appropriate scientific equipment. I can take measurements with increasing accuracy and

	and mix. I can notice when something changes during my exploration.	when I change something. I can make observations during an investigation.		compare results.	using standard units.	appropriate. I can take accurate measurements using standard units.	units. I can take repeat readings when appropriate.	precision. I can take repeat readings when appropriate and explain why these can improve the quality of my evidence.
	Recording Evidence							
	I can talk about what I notice. I can draw what I see. I can use marks, pictures or objects to show what I have found.	I can talk about, draw and record what I observe. I can use simple pictures, marks, writing or technology to record what I find. I can compare my observations.	I can gather and record simple data to help answer questions with prompting. I can record my observations using pictures, words or simple tables.	I can gather and record data to help answer questions. I can use simple scientific language when recording my findings. I can organise information using simple tables, diagrams and charts.	I can record findings using scientific language, drawings and labelled diagrams. I can record data using tables, bar charts and simple keys. I can gather, classify and present data in different ways to help answer questions.	I can independently record findings using appropriate scientific language. I can use drawings, labelled diagrams, keys, tables and bar charts to record findings. I can gather, classify and present data in ways that help me identify patterns and answer questions.	I can record data and results of increasing complexity. I can use scientific diagrams, labels, classification keys, tables, scatter graphs, bar graphs and line graphs where appropriate. I can select an appropriate way to present my data with some prompting.	I can independently select appropriate ways to record and present data and results. I can use scientific diagrams, classification keys, tables, scatter graphs, bar graphs and line graphs where appropriate. I can present data clearly so that patterns and relationships can be identified and interpreted.
	Reporting Findings							
	I can talk about what I notice.	I can describe what happened during	I can talk about what I have found out.	I can identify and classify things	I can report findings from enquiries using	I can report findings from enquiries using	I can report and present findings from enquiries in	I can independently report and present

	I can describe similarities, differences and changes. I can use talk, pictures and marks to share what I have found.	an exploration or investigation. I can talk about similarities, differences, patterns and changes. I can share what I have found using talk, pictures, writing or other forms of recording.	I can begin to identify and classify things using my observations.	more independently. I can describe what I have found using simple scientific language. I can use my results to answer a simple question.	oral and written explanations. I can describe results and simple conclusions with support. I can present findings using displays or presentations with support.	oral and written explanations. I can describe results and conclusions using appropriate scientific language. I can present findings using displays and presentations.	oral and written forms. I can identify and describe causal relationships where appropriate. I can explain my conclusions using evidence from my results. I can begin to comment on the degree of trust I have in my results.	findings from enquiries in appropriate oral and written forms. I can explain conclusions and causal relationships using scientific evidence. I can evaluate the degree of trust I have in my results. I can communicate my findings clearly using appropriate scientific language and representations.
Predictions & Conclusions								
	I can say what I think might happen. I can notice similarities, differences and changes. I can talk about whether something happened as I expected. I can suggest	I can make a simple prediction about what might happen. I can use what I observe to talk about what happened. I can identify simple patterns, similarities and differences. I can talk about	I can make a simple prediction with support. I can use my observations to suggest answers to questions.	I can make simple predictions. I can use my observations and ideas to suggest answers to questions. I can identify simple patterns in my results.	I can identify differences, similarities and changes related to simple scientific ideas and processes. I can use results to draw simple conclusions. I can make predictions for new values with	I can identify differences, similarities, changes and patterns in my results. I can use results to draw simple conclusions. I can use results to make predictions for new values. I can suggest	I can use test results to make predictions and set up further comparative and fair tests. I can identify scientific evidence that has been used to support or refute ideas or arguments with some prompting. I can identify patterns and	I can use test results to make predictions and independently plan further comparative and fair tests. I can identify scientific evidence that supports or refutes ideas or arguments. I can interpret patterns and relationships in

	something I could try differently.	whether my prediction was correct.			support. I can suggest improvements and raise further questions. I can use straightforward scientific evidence to answer questions.	improvements and raise further questions. I can use scientific evidence to support my findings.	relationships in results. I can consider whether my results are reliable and suggest improvements to an enquiry.	results to draw evidence-based conclusions. I can evaluate the reliability of my results and identify possible sources of error. I can suggest further questions and investigations arising from my findings.
Vocabulary	Observe, notice, look, listen, explore, change, same, different, pattern, try, question	Predict, observe, question, explore, test, compare, pattern, change, record, explain	Investigate, observe, question, test, compare, identify, classify, group, record, predict	Data, observe, measure, classify, sort, compare, record, table, chart, pattern, predict, conclusion	Enquiry, observation, measurement, standard unit, comparative test, fair test, evidence, data, table, bar chart, classification key, conclusion, prediction	Variable, systematic, accurate, measurement, data logger, fair test, comparative test, evidence, classify, pattern, results, conclusion, causal relationship	Precision, variable, control variable, repeat reading, accuracy, data, scatter graph, line graph, evidence, causal relationship, reliability, conclusion, comparative test, fair test, secondary source	Reliability, precision, accuracy, variable, control variable, repeat reading, evidence, scientific evidence, causal relationship, interpretation, conclusion, prediction, refute, support, degree of trust, secondary source